Genomic emergence of resistance to the BPaL/BPaLM regimen across the *Mycobacterium tuberculosis* complex: standing pocket variation versus drug-era determinants

Christophe Guyeux[1,*]

[1]FEMTO-ST Institute, DISC Department,
Univ. Marie et Louis Pasteur, CNRS UMR 6174, Besançon, France

*Corresponding author: `christophe.guyeux@univ-fcomte.fr`

Abstract

The WHO-recommended short BPaLM regimen (bedaquiline, pretomanid, linezolid, moxifloxacin) for multidrug-resistant tuberculosis depends on two drugs whose resistance is poorly catalogued: pretomanid has neither a catalogue entry nor public drug-susceptibility testing (DST) data. Using a consolidated mutation catalogue and a 132,609-strain variant pangenome of the *Mycobacterium tuberculosis* complex (MTBC), we map where, when and in which lineages BPaLM resistance determinants arise, and test whether lineages are *primed* by pre-existing polymorphisms. Raw L2/Beijing enrichment for all BPaLM resistances splits after correcting for clinical-sampling and curation bias: an artefact for bedaquiline and clofazimine (OR $\approx$1.1) but genuine for delamanid (cluster-robust OR 4.3, 95% CI 2.1–8.7), linezolid and the fluoroquinolones. Lacking DST for pretomanid, we discover F420-pathway determinants phenotype-free by phylogenetic convergence, triangulated with structural localization in the F420 cofactor pocket and ESM-1v scoring, surfacing buried, convergent candidates (*fgd1* position 210, uncatalogued at any grade and struck by five residues across four lineages; *fgd1* position 10; *ddn* position 111) that are predicted damaging but rest on structural and convergence evidence rather than genome-wide significance. An emergence-profiling test, validated against catalogued drug-era determinants (*ddn* L49P, W139*) and a catalogued-but-clonal negative control (*ddn* W88*), shows these candidates are *standing* rather than drug-era selected (most predate the drugs, are multi-clade, and sit on non-MDR backgrounds), i.e. candidate *priming* polymorphisms rather than actively selected determinants. Literature mining independently establishes the F420 pathway as a drug-inducible resistance route *in vitro* and identifies a fourth, DST-anchored candidate, *fbiC* V318I, recurrent across six sequencing projects over a decade rather than one over-sampled clone. A compensatory-background analysis, used as a negative control for the F420-standing-variation account, shows fluoroquinolone resistance riding the fit, compensated MDR clone while delamanid/F420 resistance emerges independently. This delivers a systematic account of BPaLM resistance emergence, identifying standing F420-pocket priming variation distinct from drug-era-selected determinants.

Keywords: *Mycobacterium tuberculosis*, MTBC, BPaLM, bedaquiline, pretomanid, F420 activation, drug-resistance emergence, standing variation, phylogenomics.

1 Introduction

Since its 2022 WHO recommendation [1], the all-oral short-course BPaLM regimen, combining bedaquiline (BDQ), pretomanid (PMD), linezolid (LZD) and moxifloxacin (MFX), has become the standard of care for multidrug- and rifampicin-resistant tuberculosis, validated by the Nix-TB [2], ZeNix [3] and TB-PRACTECAL [4] trials. Its efficacy rests on two new drugs whose resistance is already emerging, and recent editorials frame the same risk: without resistance diagnostics deployed in parallel, the new drugs may be lost soon after their introduction [5, 6].

Bedaquiline resistance proceeds through two routes: a rare high-level target route via *atpE* [7, 8] and a dominant efflux route via loss-of-function of *Rv0678*/*mmpR5*, the repressor of the MmpL5–MmpS5 pump [9]. The latter cross-confers clofazimine resistance, so clofazimine exposure selects for BDQ resistance [10, 11], and acquisition is typically stepwise, *Rv0678* preceding *atpE* [12]. Genomic sensitivity nonetheless remains low: novel BDQ determinants outside the canonical loci are repeatedly reported [13], and the absence of a well-defined resistance locus is now stated explicitly [14].

Pretomanid (and delamanid, DLM) are pro-drugs bioactivated by the deazaflavin-dependent nitroreductase Ddn [15], itself dependent on the cofactor F420 produced by *fgd1* [16] and the F420-biosynthesis machinery *fbiA*/*fbiB*/*fbiC* (and *fbiD*) [17, 18]. Any loss-of-function along this pathway abolishes activation and confers resistance, and because delamanid shares this exact bioactivation mechanism, pretomanid and delamanid share their genetic basis of resistance [19, 20]. The mutational spectrum was sharpened in 2026 [21, 22], the structural basis of Fgd1-mediated resistance resolved [23, 24], and a multi-country African cohort showed that phenotypic resistance is rarely observed for most WHO group-3/4 variants in these very genes [25]. For the present work, *neither the WHO 2023 catalogue [26] nor public DST sets cover pretomanid*: PMD resistance can be approached only through the delamanid/F420 bridge, convergence and functional prediction.

A third thread is that of pre-existing, or *standing*, resistance variation. This is not a new idea: natural polymorphisms in the F420 genes confer delamanid resistance in drug-naive patients [27], the genetic diversity of the BPaL candidate loci has been mapped across tens of thousands of genomes [28], the intrinsic susceptibility of the MTBC to pretomanid differs by lineage in both ancient and recent ways [29, 30], bedaquiline-candidate variants pre-exist in bedaquiline-naive patients and track lineage [31], and a historic reservoir of bedaquiline/clofazimine resistance-associated variants predates the drug [32, 33]; the natural diversity of *fgd1*/*ddn* was already shown to modulate PA-824 susceptibility a decade before pretomanid was approved [34], and widespread pre-existing loss-of-function has been quantified at genome scale [35]. A pre-resistance framework, reconstructing ancestral states on a timed tree to measure the hazard of acquiring resistance from standing polymorphisms, is likewise established [36], and the baseline-versus-acquired distinction has been operationalized on the pretomanid trials themselves [37]. What is missing, and what this study adds, is a single cross-lineage genomic analysis that (i) *dissociates* benign lineage-defining polymorphism from functional pocket variation by a structural filter (distance to the F420 cofactor), (ii) separates standing/priming variation from drug-era-emergent determinants by an emergence-recency test calibrated against catalogued anchors, and (iii) extends the historic-reservoir / pre-resistance reading, so far applied to the bedaquiline efflux locus [33], to the F420 activation pathway, where it has never been done. Intrinsic, lineage-dependent tolerance provides the phenotypic backdrop [38–40].

We therefore ask, across the MTBC: (i) where and in which lineages catalogued BPaLM determinants are prevalent, once sampling and curation bias are controlled; (ii) which lineages carry pre-existing, lineage-fixed polymorphisms in the BDQ/PMD machinery, and whether these are functional; (iii) which *novel* F420-pathway determinants can be discovered phenotype-free by convergence, given the pretomanid catalogue gap; (iv) whether such candidates are drug-era emergent or standing/priming variation; and (v) whether a compensatory MDR background facilitates emergence of the new drugs. The study is deliberately positioned as an evolutionary/epidemiological account that *consumes* a mature resistance-catalogue infrastructure rather than re-deriving it.

2 Materials and Methods

2.1 Data and reference resources

All analyses consume, without re-deriving, a consolidated antimicrobial-resistance reference infrastructure: a mutation-to-resistance catalogue reconciling the WHO catalogue (2nd ed., 2023) [26], the tb-profiler reference database and empirical CRyPTIC signal (62,020 graded assertions over 21 drugs); a strain $\times$ variant pangenome of 132,609 MTBC genomes in 0-based SPDI coordinates, built on the TB-Annotator platform [41]; a phenotype-consensus table of drug-susceptibility calls; and a Coll-scheme lineage classification of 248,771 strains [42] (major human lineages `L1`–`L7` and the animal ecotypes `BOV` and `BOV_AFRI`). Variants are written as SPDI (`NC_000962.3:position:ref:alt`) against H37Rv. Per-BioProject geographic origin and collection date were taken from a consolidated geolocation table covering 146,216 strains (67,312 of them present and dated within the pangenome). A 15-gene panel (Table 2) was extracted from the H37Rv GFF3 annotation. The strain cohorts used per analysis are summarised in Table 1.

Table 1: Strain cohorts used in this study.

Cohort	Size	Used in
Variant pangenome	132,609 strains	frequency / lineage-prevalence analyses
dated subset	67,312	emergence-recency test (background)
Delamanid DST	187 R / 11,739 S	DST validation (F420 phenotype proxy)
Rifampicin DST	10,920 R / 41,232 S	compensatory-background analysis

2.2 Feasibility scan and trichotomy

A deterministic codon-effect annotator (validated on the *gyrA* D94 fluoroquinolone series) scanned all panel positions across the pangenome, cross-referenced each variant to the catalogue, and yielded 22,806 distinct panel variants. Every variant was classified by a strict *trichotomy* that operationalizes the "a variant in a DR gene is not a resistance variant" guard: *catalogued-R* (graded R-associated at a confirmed, DST-supported grade; WHO interim/uncertain-significance calls with zero empirical PPV are not treated as catalogued-R here, see Discussion), *lineage-defining polymorphism* (non-synonymous, not catalogued-R, frequency >2%, spread across lineages), or *novel candidate* (non-synonymous, not catalogued-R). Because the dominant *Rv0678* mechanism is indel-mediated, BDQ counts were reconciled at the amino-acid level (SNV+MNV) rather than from SNV codons alone.

2.3 Lineage stratification and bias correction

Catalogued-R prevalence was computed per major lineage with Clopper–Pearson 95% CIs and Fisher tests (L2 vs. rest). Because raw prevalence is confounded by over-representation of MDR clinical cohorts and by heterogeneous per-lineage curation, we applied two reliable adjustments: comparison against an *equally-curated* comparator (L2 vs. L3), and a BioProject cluster-robust logistic regression that controls pseudo-replication. (Equal-weight inverse-probability weighting by BioProject was found unstable, since it over-amplifies small resistant-enriched studies, and is reported only as a sensitivity check.)

2.4 Primed polymorphisms and functional prediction

A polymorphism was deemed *primed* (pre-dating the drug era) if non-synonymous, not catalogued-R, concentrated $\geq$85% within one lineage and present in $\geq$5% of it (i.e. fixed on the lineage trunk, hence ancestral to drug introduction). Functional impact was scored by the ESM-1v zero-shot masked-marginal log-likelihood ratio (LLR; Meier et al. [2021]); the scale was calibrated by controls (known damaging *katG* S315T -7.52, *rpoB* S450L -9.63; tolerated lineage marker *gyrA* E21Q $+2.84$). LLR $\ll 0$ denotes a disfavoured/damaging substitution. The LLR is used as a qualitative prior, anchored by these controls, not as a calibrated probability.

2.5 Structural localization

Candidate residues were localized on holo crystal structures carrying the F420 cofactor (PDB ligand F42): Fgd1 = 3B4Y (1.95 Å, with citrate FLC marking the active centre) and Ddn = 3R5W. Atom–atom distances to F420 and relative solvent accessibility (RSA; Shrake–Rupley via Biopython [44], normalized per Tien et al. [2013]) were computed; thresholds <4 Å direct contact, 4–8 Å first shell, 8–12 Å second shell, >12 Å distal; RSA <0.20 buried, >0.40 exposed.

2.6 Phenotype-free convergence discovery

Independent recurrence of non-catalogued non-synonymous variants of the BDQ/PMD/F420 genes was scored across major lineages (collapsing each lineage to one unit; a within- lineage frequency ceiling of 10% separates recurrent emergence from fixed lineage markers). The detector was positive-controlled on *gyrA*/*gyrB* (it must recover the known fluoroquinolone QRDR sites before novel F420 hits are interpreted). Positional convergence (several alternative residues at one site) is the strongest signal; homoplasy is read as a signature of positive selection following established practice [46].

2.7 Phylogenetic test of independent origins

To verify that the convergence is genuine rather than an artefact of collapsing strains to major lineages, we built a maximum-likelihood tree of 969 strains (all candidate and anchor carriers, a 50-strain-per-lineage backbone, and a positive control of *gyrA* A90 carriers) from 10,000 parsimony-informative binary SPDI sites (RAxML-NG, BIN+GΓ). The presence/absence of each candidate variant was reconstructed on the tree by Fitch parsimony, and the number of independent 0→1 transitions (independent origins, i.e. mapped homoplasy) was counted [46]. The count was calibrated by a homoplastic positive control (*gyrA* A90) and a clonal negative control (the *ddn* W88* L2-MDR clone marker). To assess whether the positional multiplicity of alternative residues exceeds chance given the mutational density of each gene, we also ran a per-gene permutation null (20,000 replicates) randomly placing each non-synonymous variant on a codon, recording how often a single codon reached the observed multiplicity.

2.8 DST validation and emergence profiling

Convergent F420 candidates were confronted with the delamanid cohort of the consensus (187 R / 11,739 S; DLM is the F420 phenotype proxy for PMD) by direct carrier lookup, Fisher case–control on all F420-gene variants (positive control: catalogued DLM-R determinants), and a reverse position histogram among the resistant strains. Emergence was profiled over the full pangenome per candidate position: collection-year distribution, major-lineage and sub-lineage spread, country, and co-occurrence of a catalogued rifampicin-R determinant (MDR background). A *recency-vs-background* test (one-sided Mann–Whitney of carrier years against all dated pangenome strains, median 2015) asks whether carriers are more recent than the sampling trend; it was internally validated against catalogued DLM-R anchors. To control for geographic sampling artefacts, the test was repeated against lineage-matched and country-matched backgrounds (restricting the comparator to the major lineages, then the countries, of the carriers).

2.9 Literature-anchored screening and clonality control

Body-text DST tables from 51 tables across 16 recent BPaL/BPaLM papers were mined for F420-pathway variants with a reported phenotype (Methods, litcohort pipeline). Eleven positions phenotyped in the literature but absent from our convergent candidate set were screened for carriers in the full pangenome by direct position lookup. For a resulting candidate with a substantial carrier count (*fbiC* V318I, $n = 10$), clonality was ruled out before treating the carriers as independent evidence: pairwise genome-wide SNP distances were computed on the union of variable positions among the 10 carriers, against the standard 12-SNP recent-transmission threshold [47].

2.10 Compensatory background

As a negative control for the F420-standing-variation account above, testing whether a fit, compensated MDR background could instead epidemiologically explain BPaLM resistance, compensatory *rpoC*/*rpoA* variants were identified empirically as those enriched among rifampicin-R strains (Fisher carrier × RIF phenotype, cohort 10,920 R / 41,232 S; OR $\geq$3, $p < 10^{-2}$), cross-checked against known positions. BPaLM resistance was then compared between compensated and non-compensated strains, both unconditioned and *restricted to RIF-R* (i.e. at equal MDR status).

2.11 Generative AI use in the analysis workflow

The bioinformatic analyses, database queries and literature cross-referencing underlying this study were carried out within an AI-augmented bioinformatics working environment, Claude Code (Anthropic), operating the Claude Sonnet 5 language model as an interactive analysis agent under continuous human direction. Every script, statistical test and numerical claim produced within this environment was independently traced back to its underlying raw data, primary publication or database record before being reported here, following the verification discipline described throughout this section; the corresponding author designed the study, specified and reviewed every analysis, and takes full responsibility for the accuracy of its content.

3 Results

3.1 The substrate supports the study, and the trichotomy is operational

The feasibility scan returned a clean positive control (the *gyrA* D94 series recovered: D94G×99, D94N×37, D94Y×37, D94A×21, D94H×14). *Rv0678* carried 550 variants, 79 catalogued-R with 895 carriers, the top carriers being indels/frameshifts as expected. *ddn* carried 24 catalogued-R variants, all truncations (W88*, W139*, W20*, Q58*) or L49P, i.e. loss-of-function of the activating nitroreductase. The F420 pathway carried 2,167 non-synonymous variants, of which 1,696 were non-catalogued, the discovery reservoir that motivates the project, and a roughly four-fold larger substrate than the previous genome-scale survey of these loci [28]. A large lineage confounder is present (*gyrA* E21Q in 93% of strains, G668D 85%, S95T 81%; *rpoC* 122k carriers; a synonymous *fgd1* F320F in 68k), confirming that the trichotomy is indispensable: overall, 157 catalogued-R SPDI, 13 lineage-defining polymorphisms, and 4,493 non-catalogued candidates (1,696 in the F420 pathway).

3.2 Bias correction splits a uniform Beijing enrichment

Raw prevalence (Fig. 1) made L2/Beijing look enriched for *all* BPaLM resistances (L2 vs. rest: DLM OR 4.4, $p = 10^{-37}$; LZD 5.3; FQ 3.25; BDQ 1.8; CFZ 1.7). After bias correction (Fig. 2), the picture splits: for bedaquiline and clofazimine the enrichment is an artefact, since it collapses against the equally-curated L3 comparator (OR 1.14) and is non-significant in the cluster-robust model (95% CI includes 1), consistent with sporadic, under-treatment efflux acquisition rather than a lineage trait. For delamanid, linezolid and the fluoroquinolones the enrichment is genuine: delamanid is the most marked (cluster-robust OR 4.27, 95% CI 2.10–8.67, over 58 BioProjects). Because delamanid and pretomanid share F420 activation, a real Beijing enrichment for delamanid is directly relevant to pretomanid.

3.3 The F420 machinery carries lineage-fixed standing variation, but the surface markers are benign

Eleven non-catalogued F420 variants are fixed within single lineages (Table 3): *ddn* D113N fixed in L5 (99.6%), *fgd1* K296E in L6 (98.1%), *fgd1* K270M in an L4 sub-lineage (22% of L4), *fgd1* R64S in L1 (11.6%), *ddn* L90V in BOV_AFRI (51%), plus *fbiB*/*fbiC*/*fbiD* variants. ESM-1v flags several as damaging (*fgd1* R64S −8.33, as severe as *katG* S315T). However, holo-structure localization *refutes* a functional role for the *fgd1* trio (Fig. 4A): all four candidate primers are distal from the F420 cofactor (R64 21.4 Å, K270 18.6 Å, K296 17.2 Å, exposed/surface; D113 12.1 Å, second shell), whereas the catalytic landmark (citrate) sits 3.3 Å from F420. Combined with relaxed purifying selection on *fgd1* (pN/pS 0.66) versus purifying selection on *ddn* (pN/pS 0.20), the parsimonious reading is that the *fgd1* surface variants are family-disfavoured but neutrally fixed *lineage markers*, not functional primers; this corrects an earlier lean that had proposed *fgd1* R64S as the mechanism of L1 low pretomanid susceptibility. L2/Beijing carries *no* fixed F420 primer, so its delamanid enrichment is *acquired* under pressure, not intrinsic priming, two distinct mechanisms that must not be conflated.

3.4 Phenotype-free convergence surfaces buried cofactor-pocket candidates

The convergence detector recovered the known fluoroquinolone signal as a positive control (Fig. 4B; *gyrA* A90 struck by E/G/S/V across 7 lineages, 3,162 carriers; the *gyrB* QRDR). In the F420 activation pathway it surfaced *positional* convergence: *fgd1* position 210 struck by five residues (A210 E/G/P/T/V) across four human lineages (L1–L4), entirely non-catalogued, and *fgd1* position 10 (A10 V/P/T; A10V already *delamanid:uncertain*). Three independent lines of evidence coincide (Table 4, Fig. 4A): positional multi-lineage convergence; first-shell, buried localization of the F420 pocket (fgd1 10 at 7.6 Å/RSA 0.00, fgd1 210 at 7.8 Å/RSA 0.04, versus the distal lineage-fixed markers at 17–21 Å); and strong ESM-1v damaging scores (the five fgd1 210 substitutions from −5.60 to −8.39, bracketing *katG* S315T). These are the distinctive phenotype-free PMD/DLM candidates. For BDQ, convergence recovered a cloud of recurrent non-catalogued *Rv0678* missense (R50Q across 5 lineages) consistent with catalogue extension at a known hotspot.

A maximum-likelihood phylogeny of 969 strains confirms, and strengthens, this convergence (Methods; calibrated by the positive control *gyrA* A90 reconstructed to 28 independent origins and the negative control *ddn* W88*, a clonal L2-MDR marker, to 2). The candidate positions arose by multiple independent origins on the tree: *fgd1* 210 by six, *fgd1* 10 by four and *ddn* 111 by eight independent 0→1 transitions (versus 9, 9 and 38 carriers respectively). The phylogeny thus does not inflate the convergence: it recovers *more* independent origins than the major-lineage collapse of Phase 5 (which reported four lineages each), because several origins occur within a single major lineage. The catalogued anchor *ddn* L49P, by contrast, traces to a single origin (57 carriers), confirming that origin count and frequency are decoupled. The origin count is also the symmetric clonal-linkage control for the retained candidates: their four to eight independent origins exclude a single-transmission-cluster artefact, the same criterion under which the clonal *ddn* Y65S (one origin) is demoted below.

3.5 DST validation redirects, then emergence profiling reframes

On the delamanid cohort, the method validated cleanly (catalogued *ddn* L49P 7R/1S, OR 456, $p = 1.6 \times 10^{-12}$; W88*/W139* truncations 2R/0S), and the catalogued L49P itself localizes to the first shell (7.7 Å, buried), anchoring the "pocket = functional" criterion on a graded determinant. The convergent *fgd1* 210/10 variants, however, are *DST-orphan* (a single carrier tested, A10V→S): standing/sporadic variants are nearly absent from MDR-selected clinical DST cohorts, which is precisely why phenotype-free discovery is necessary. Case–control surfaced new *ddn* pocket candidates (Y65S 2R/3S, OR 42, 3.1 Å direct F420 contact; G81S 2R/0S, 7.0 Å).

Literature mining closes part of this DST gap and adds a fourth, epistemically distinct target. Of 51 body-text DST tables screened, only one links drug and accession-level phenotype (the delamanid/clofazimine cohort already used above); the anchor *ddn* L49P is nonetheless independently confirmed as pretomanid-resistant, not only delamanid-catalogued, in two isogenic constructions (MIC >16, primary high-level resistance [48]; MIC 30 mg/L, 300-fold wild-type [49]), and the causal reach of the F420 pathway itself is now established directly *in vitro*: serial clofazimine exposure of H37Rv selects *fbiA*/*fbiC*/*fbiD* mutants conferring de novo delamanid and pretomanid resistance, with no lineage or clinical-sampling confound [50]. Screening our substrate at the 11 literature-phenotyped positions absent from our convergent set recovers carriers at 7/11; two match a reported resistant genotype exactly, *ddn* Y29* (already catalogued delamanid:R-associated) and *fbiC* V318I ($n = 10$, all L2, matching a delamanid-resistant isolate from an Iranian MDR cohort [51]). Because a recurring genotype could reflect one over-sampled clone rather than independent carriage, genome-wide SNP distances among the 10 *fbiC* V318I carriers were checked: they span six BioProjects, 2013–2023, and two countries, with only 2 of 45 pairs falling under the 12-SNP transmission threshold [47] (one pair at distance 0, plausibly a technical duplicate; one pair at exactly 12, sharing a BioProject) and a median pairwise distance of 44 SNP – consistent with roughly seven to eight distinct genetic backgrounds, not a single clone. *fbiC* V318I is therefore retained as a DST-anchored, non-clonal fourth target, distinct from *fgd1* 210/10 and *ddn* 111 in being literature-phenotyped rather than phenotype-free, but lacking their experimental structure (*fbiC*, Rv1173, has no solved PDB structure, so it cannot be localized in the F420 pocket). A fifth literature-flagged position, *fbiA* R321S, is not retained: its single Iranian carrier co-occurs with five other F420/ribosomal variants on a hyper-mutated MDR background, and the two catalogue-supporting isolates are themselves untraceable, so the co-occurrence confound cannot be excluded.

Emergence profiling (Fig. 3; Table 4) then reframes the whole candidate set. The recency-vs-background test is internally valid: it lights up on the genuinely drug-era catalogued anchors (*ddn* L49P median 2018, $p = 0.010$; W139* median 2020, $p = 0.032$) and stays flat on a catalogued-but-clonal one (*ddn* W88* median 2011, $p = 0.99$, a Georgia/Moldova L2-MDR clone marker, 96% RIF-R). Against this calibrated test, none of the convergent candidates is more recent than the sampling background (all $p \geq 0.17$), and this non-recency is robust to lineage- and country-matched backgrounds (all $p \geq 0.22$ and ≥ 0.43 respectively); they carry firm pre-2014 carriers (*fgd1* 10: 2; *ddn* 111: 8; *ddn* 81: 56), proving they pre-date delamanid/pretomanid clinical use (collection date $\leq$ acquisition date). They are genuinely multi-clade (*fgd1* 210 across 4 major lineages / 6 sub-lineages; *ddn* 111 across 4/7) and sit on predominantly non-MDR backgrounds (RIF-R co-occurrence 0–22%). This is the profile expected of *standing, priming* variation, not of determinants under active drug-era selection. The exception is *fgd1* 210, whose six dated carriers include none from before 2014: it is contemporary with, rather than clearly antecedent to, the sampling background (recency $p = 0.17$, under-powered), so its standing status is suggestive but not established. The same lineage-spread lens demotes the two phenotype hits: *ddn* Y65S is clonal (10 carriers, all L4, one sub-lineage; its OR 42 is lineage-linkage confounding), and *ddn*

G81S is an L2-Beijing standing polymorphism on the MDR background (99/101 L2, 66% RIF-R).

3.6 Compensation carries the fluoroquinolones, not the F420 drugs

Empirically-identified compensatory *rpoC*/*rpoA* variants (84 variants, recovering textbook hotspots: *rpoC* V483G 522R/5S, OR 414; I491V; V483A) define 21,309 compensated strains, 89% RIF-R and 69% L2. Crude associations of BPaLM resistance with the compensated background are strong but, conditioned on RIF-R (i.e. at equal MDR status), they collapse except for the fluoroquinolones (levofloxacin OR 6.30→1.89, $p = 5.8 \times 10^{-26}$; moxifloxacin 5.34→1.55, $p = 2.5 \times 10^{-14}$), with bedaquiline marginal (3.20→1.44, $p = 0.036$) and delamanid/linezolid non-enriched (OR <1). Fluoroquinolone resistance genuinely accumulates on the fit, compensated MDR clone (a pre-XDR trajectory on a successful backbone), whereas delamanid/F420 resistance emerges *independently* of the compensation–*rpoB* axis, consistent with the F420 results above. This negative result is itself informative: had DLM/F420 resistance instead tracked the compensated MDR background, the standing-variation account above would have been confounded by clonal linkage to a successful lineage; its absence strengthens, rather than merely accompanies, the main finding.

4 Discussion

A single, coherent picture emerges in two regimes. The *drug-era emergent determinants* are the catalogued *ddn* loss-of-function variants (L49P, W139*), recent and selected, exactly as expected. The *standing, priming pocket variation* is the convergent F420 set (*fgd1* 210/10, *ddn* 111): multi-clade, cofactor-pocket, ESM-damaging, on non-MDR backgrounds, and pre-dating the drug era. The trio is structurally heterogeneous rather than a uniform block, however: *fgd1* 210/10 sit in the first shell (7.6–7.8 Å, RSA ≈ 0, ESM-1v down to −8.4), whereas *ddn* 111, despite carrying the strongest empirical signal of the three (38 carriers, eight independent origins), is second-shell (13 Å) with a roughly twofold weaker ESM-1v effect (−5.0; Table 4). This is consistent with a distinct mechanism, a fold-stability contribution rather than direct cofactor contact, and does not change *ddn* 111's status as a prioritized phenotype-free target, but the trio should not be read as mechanistically uniform. The two poles are separated by an explicit structural filter, the distance to the F420 cofactor on holo structures, which distinguishes benign lineage markers (distal, surface) from buried pocket candidates, and which is empirically anchored on the catalogued L49P determinant. This reframes the project's contribution: not "new determinants under active selection" (the temporal profile does not support that), but the systematic identification of *priming* standing variation in the F420 pocket, shared by convergence across lineages.

This must be positioned carefully against an established prior art, because both poles of the dichotomy are individually pre-described. Standing variation in the BPaL candidate loci is documented [27, 28, 31, 35], lineage-dependent intrinsic pretomanid susceptibility is established [29, 30], and the drug-era-emergent pole, *ddn* loss-of-function, is also known. Critically, a *ddn* loss-of-function determinant is *not* automatically drug-era emergent: the *ddn* Trp20Stop and in-frame deletions are clade-restricted to L4.5 and confer *primary* (pre-treatment) high-level resistance [48, 52]. The standing-versus-emergent axis is therefore not "F420 missense = standing, *ddn* LoF = emergent"; it is an axis of evolutionary signature (multi-clade convergence on a non-MDR background with pre-drug presence, versus clade-restricted or clinically-selected appearance), which is why the recency-vs-background test and the lineage-spread lens are needed rather than the mechanism alone. Our convergent, multi-clade *ddn* 111 is thus distinct from the clade-restricted L4.5 *ddn* loss-of-function, and our increment over the genome-scale standing-diversity maps [28, 35] is the explicit dissociation by structural pocket localization, ESM-1v per-position effect, and emergence-recency calibration, extending the historic-reservoir reading from the efflux locus [33] to the F420 activation pathway. The mechanism by which a standing pocket variant can later become a clinical determinant, selection of pre-existing mutants under sub-inhibitory exposure, is itself documented [53].

The candidate positions are phenotype-free predictions, and the published functional record supports their plausibility at the gene, pathway and pocket level without yet confirming the individual residues. Isogenic and spontaneous-mutant studies establish that *ddn*, *fgd1* and the F420-biosynthesis genes are bona fide pretomanid/delamanid resistance determinants [21, 34, 54, 55], and the functional importance of the F420-binding region is established structurally and by mutagenesis [23, 24, 56–58]; notably, neither independent structure-guided screen [23, 24] flags residues 210 or 10 of *fgd1*, underscoring how sparsely the pocket's mutational landscape has been sampled outside catalogue-driven positions.

A codon-level audit of the WHO catalogue against the three convergent positions corrects our earlier, looser framing of "novelty". Only *fgd1* 210 is genuinely absent from the catalogue at any grade. *ddn* 111 carries an interim, mechanism-only grade (*Ala111fs*, group 2 "Assoc w R – Interim", frameshift ⇒ presumed loss-of-function), and *fgd1* 10 an uncertain-significance grade (*Ala10Val*, group 3); both grades are empirically unsupported (PPV = 0.0, one tested isolate each), a rule-based extrapolation rather than a DST-confirmed call. Being listed is therefore not the same as being explained: neither catalogue entry carries a multi-lineage, pre-drug-era, structurally-localized reading, which is what this analysis adds. We accordingly present *fgd1* 210 as a genuinely uncatalogued phenotype-free target, and *ddn* 111 / *fgd1* 10 as catalogued-but-mechanistically-unexplained standing candidates; in both cases the contribution is the evolutionary and structural context, not the listing status. *ddn* Y65, by contrast, coincides with an active-site residue already characterized by mutagenesis and molecular dynamics [56, 57], although its clonal L4 distribution (above) precludes a convergence claim. We therefore present *fgd1* 210/10 and *ddn* 111 as *prioritized phenotype-free targets* for confirmatory susceptibility testing, not as established determinants. Literature-anchored screening (above) adds a fourth target of a different epistemic kind, *fbiC* V318I: not phenotype-free (it matches a delamanid-resistant Iranian isolate [51]) and cleared of the single-clone artefact that would otherwise undermine it, but unlocalizable in the pocket for want of a solved *fbiC* structure. The two target classes are complementary rather than redundant: the convergent trio is supported by structure and multi-origin phylogeny but awaits DST, while *fbiC* V318I is DST-anchored but awaits structure.

The clofazimine cross-selection result [50] demonstrates that the F420 pathway is a real, drug-inducible resistance route, which strengthens the premise of the whole discovery strategy; but the mechanism it describes leaves no detectable trace in our clinical cohort. Testing the epidemiological prediction of that in vitro route directly, an F420-gene variant does not enrich for clofazimine resistance in our substrate (OR 0.80 overall, $p = 0.043$; OR 0.97 within L2, $p = 0.93$), even though clofazimine and delamanid resistance are themselves strongly correlated at the strain level (OR 8.74, $p = 9.7 \times 10^{-25}$). The cross-resistance route is therefore real *in vitro* but is not the explanation for the clinical clofazimine–delamanid correlation, which more plausibly reflects shared clinical exposure or co-selection on the same MDR clone – a distinction that matters because it keeps the acquired, L2-specific delamanid enrichment (above) mechanistically separate from the artefactual bedaquiline/clofazimine one, rather than collapsing them through a shared F420 pathway.

This reframing is independently corroborated, rather than undermined, by a cohort study testing the very class of variant our candidates belong to: among 89 isolates carrying 60 unique DLM/PMD-related WHO group-3/4 variants from the multi-country DIAMA cohort (Central and West Africa), phenotypic resistance above the critical concentration was rarely observed [25]. This does not test our specific residues, but it supports the premise on which the standing/priming reading rests: the F420 pocket carries abundant WHO-uncertain polymorphism that is predominantly non-causal, so a catalogue listing at grade 3/4 is weak evidence either way, and a phenotype-free, structurally-triangulated prioritization such as ours remains the more informative lens on this variation, not a substitute for confirmatory testing of it.

The lineage architecture is itself dual. L2/Beijing is *acquired*-enriched for delamanid (robust after bias correction) yet carries no fixed F420 primer, whereas L1/L5/L6 carry lineage-fixed F420 variants that are candidates for *intrinsic* priming, although structural localization shows that the most eye-catching of these (the *fgd1* surface trio) are benign, a result that enforces the structural filter. Any lineage-level reading must moreover be resolved at the sub-clade scale, since both the epidemic success of Beijing and the transmission fitness of MDR clones are sub-clade- and lineage-specific rather than uniform [59, 60]. Methodologically, the recency-vs-background test is a transferable tool: it controls the pervasive "everything looks recent because sampling is recent" confounder and, validated against catalogued anchors, lets a negative (standing) result be trusted rather than dismissed as low power. Stratification exposes its boundary: the anchors' recency is robust to lineage-matching but attenuates under country-matching (their drug-era signal is partly geographic), so the test is best read as recency relative to the lineage background. Crucially, this confound can only *inflate* apparent recency, so it cannot manufacture the candidates' standing status, which shows no recency under any background.

Limitations. Collection date bounds acquisition from above, so the "pre-2014 = standing" reading is solid while "recent = emergent" is weaker; the independent-origin counts are established on the maximum-likelihood phylogeny, but node-level *dating* of antecedence is not attempted, because the MTBC molecular clock is weak and a time-scaled tree of this focused, sparsely-dated set would be unreliable; antecedence therefore rests on the recency test and the pre-2014 carriers. Convergent candidate counts are small (*fgd1* 210/10 ≈9 carriers), and *fgd1* 210 in particular lacks pre-2014 carriers and is under-powered. The phenotype-free candidates remain predictions: pretomanid has no DST and delamanid DST is sparse, so confirmation awaits denser phenotyping. This scarcity is a measured constraint, not an assumption: of 51 body-text DST tables mined from recent BPaL literature, only one links a mutation to both drug and strain accession, so a per-strain DST reconciliation of our own carriers with published phenotypes is not achievable from the literature at present, and isogenic construction remains the only route to closing the position-level gap. Lineage and curation confounders are controlled but not fully eliminated, and all p-values are nominal (only *ddn* L49P survives genome-wide correction; novel candidates are elevated by the structural and multi-lineage prior, not by genome-wide significance). Indeed, the F420 genes are heavily polymorphic (e.g. 190 non-synonymous variants over the 151 codons of *ddn*), so the positional multiplicity of alternative residues is not exceptional on its own: a permutation null shows a single codon reaching five alternative residues in 42% of replicates for *fgd1* (and three is near-certain), so the candidates rest on the triangulation of recurrence (multi-origin, non-clonal on the tree), buried pocket localization and ESM-1v effect, not on convergence significance *per se*. The compensatory associations are epidemiological, not mechanistic.

5 Conclusion

Across 132,609 MTBC genomes, BPaLM resistance emergence resolves into an acquired, clinically-selected component (efflux BDQ, sporadic; fluoroquinolones riding the fit compensated MDR clone; delamanid acquired in Beijing) and a standing, pre-existing component of priming polymorphisms in the F420 cofactor pocket, shared by convergence across lineages and distinct from the drug-era loss-of-function determinants. Because pretomanid has neither a catalogue nor DST, this phenotype-free, structurally-filtered route is the only one available, and it is delivered here with explicit limitations. The priming candidates (*fgd1* 210/10, *ddn* pocket residues), together with the DST-anchored, non-clonal *fbiC* V318I recovered by literature-guided screening, are the natural targets for confirmatory drug-susceptibility testing, formal ancestral dating [36], and an assessment of their epidemic success [61], and for inclusion in the next iteration of the pretomanid resistance catalogue.

Data, Metadata, and Code Availability

The phased analysis scripts and the corresponding result tables, including the per-candidate carrier table (accession, collection year, country, lineage, RIF-R status) underlying Table 4 and Fig. 3, are prepared for deposit on Zenodo under an open license, with a reserved DOI to be made public upon submission: `https://doi.org/10.5281/zenodo.21993326`. This project consumes, as a read-only upstream input, a shared antimicrobial-resistance reference infrastructure (mutation catalogue, 132,609-strain variant pangenome, phenotype-consensus table) maintained by a separate internal project with its own publication and deposit timeline; that infrastructure is not redistributed in the present deposit, which documents the point explicitly.

Table 2: BPaLM drug panel and resistance genes (H37Rv loci).

Drug / role	Genes (locus)
Bedaquiline	*Rv0678*/mmpR5, *atpE* (Rv1305), *pepQ* (Rv2535c)
Pretomanide (F420)	*ddn* (Rv3547), *fgd1* (Rv0407), *fbiA* (Rv3261), *fbiB* (Rv3262), *fbiC* (Rv1173), *fbiD* (Rv2983)
Linezolid	*rrl* (23S rRNA), *rplC* (Rv0701)
Moxifloxacin	*gyrA* (Rv0006), *gyrB* (Rv0005)
Compensatory	*rpoC* (Rv0668), *rpoA* (Rv3457c)

Table 3: Lineage-fixed (primed) F420 polymorphisms, ESM-1v variant effect and structural localization. Distances are atom–atom to the F420 cofactor on holo structures (Fgd1 3B4Y, Ddn 3R5W). The damaging-but-distal *fgd1* variants are read as benign lineage markers (see Results).

Gene	Variant	Lineage (freq)	ESM-1v LLR	dist. F420	Verdict
ddn	D113N	L5 (99.6%)	−2.94	12.1 Å (2nd)	weak lead (purifying gene)
fgd1	K296E	L6 (98.1%)	−2.96	17.2 Å (distal)	benign marker
fgd1	K270M	L4 sub (22%)	−5.69	18.6 Å (distal)	benign marker
fgd1	R64S	L1 (11.6%)	−8.33	21.4 Å (distal)	benign marker
ddn	L90V	BOV_AFRI (51%)	−4.69	—	animal lineage

Table 4: Phenotype-free convergent F420 candidates and their emergence profile. "maj/sub" = distinct major lineages / sub-lineages among carriers; "pre-14" = carriers collected before 2014; "recency p" = one-sided Mann–Whitney vs. the dated-pangenome background; "RIF-R" = fraction on an MDR (catalogued rifampicin-R) background. Anchors ($\star$) are catalogued delamanid-R determinants; rows without $\star$ are phenotype-free candidates, except that a codon-level catalogue audit found that *ddn* 111 and *fgd1* 10 (†) already carry a WHO grade (interim/uncertain, PPV = 0.0 on a single tested isolate each) unsupported by DST; only *fgd1* 210 is genuinely absent from the catalogue at any grade. "—" indicates a value not determined (no holo structure for the residue, or no ESM score for a truncation). Convergent candidates are multi-clade, buried in the F420 pocket and ESM-damaging, yet *not* more recent than background (standing/priming); the anchors L49P/W139* are genuinely drug-era recent.

Position	maj/sub	dist. F420	ESM-1v	pre-14	recency p	RIF-R	Verdict
fgd1 210	4/6	7.8 Å	−8.4	0	0.17	22%	standing pocket (under-powered)
fgd1 10 †	4/4	7.6 Å	−8.2	2	0.82	0%	standing pocket
ddn 111 †	4/7	13 Å	−5.0	8	0.87	8%	standing (2nd shell)
ddn 65 (Y65S)	1/1	3.1 Å	—	3	0.84	20%	clonal L4, demoted
ddn 81 (G81S)	3/6	7.0 Å	—	56	1.00	66%	L2-MDR confounded, demoted
$\star$*ddn* 49 (L49P)	1/2	7.7 Å	—	12	**0.010**	9%	drug-era emergent
$\star$*ddn* 88 (W88*)	3/5	—	—	77	0.99	96%	clonal L2-MDR
$\star$*ddn* 139 (W139*)	2/2	—	—	2	**0.032**	5%	drug-era emergent

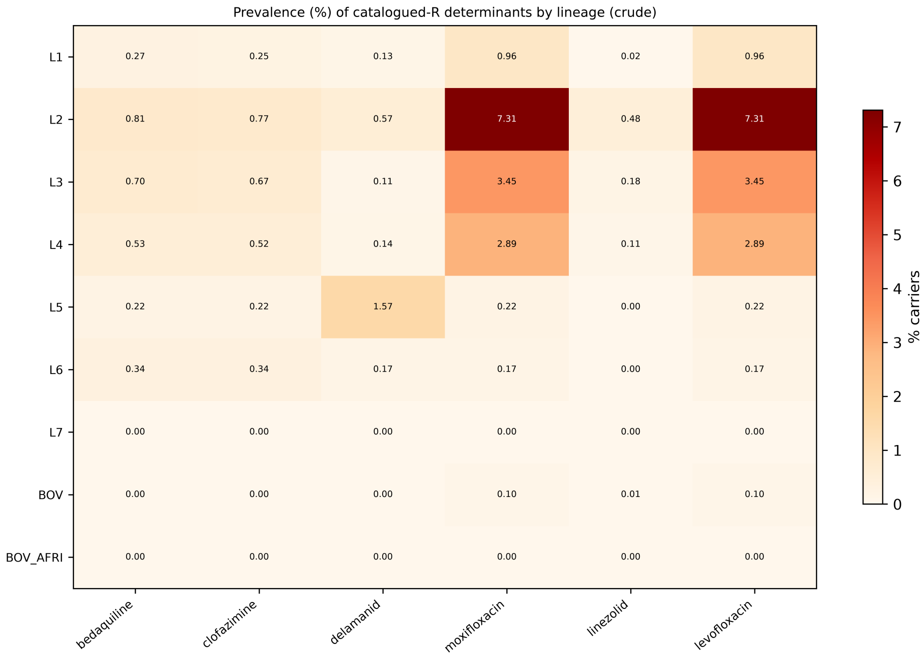

Figure 1: Raw prevalence (%) of catalogued-R determinants per major lineage and drug (132,609 strains). Values are uncorrected and must not be read as lineage traits before bias correction: L2/Beijing appears enriched for all BPaLM resistances, but this signal splits into artefact and genuine components once corrected (Fig. 2).

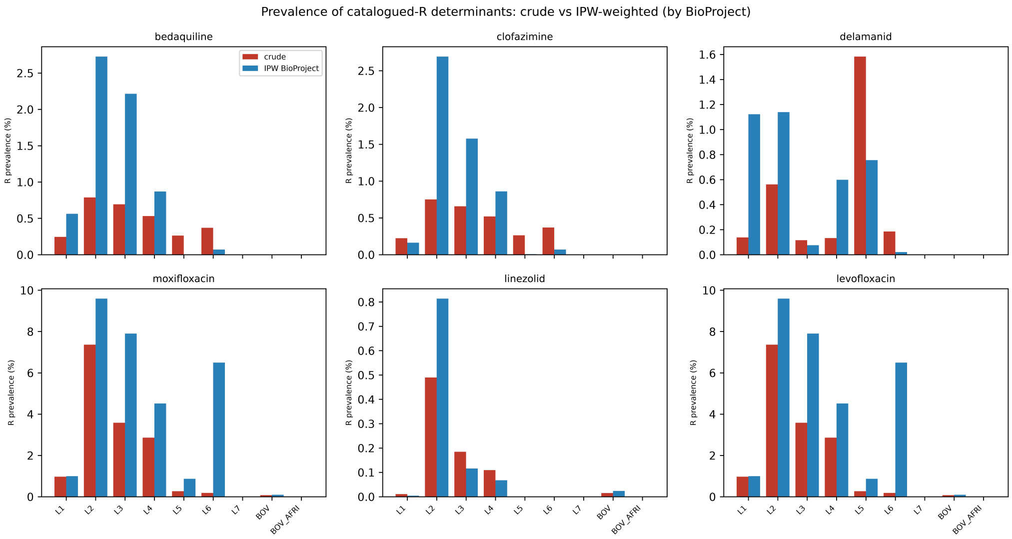

Figure 2: Crude versus bias-corrected lineage enrichment (`L2` vs. rest). After correction, bedaquiline and clofazimine enrichment is an artefact (CI includes 1), while delamanid, linezolid and the fluoroquinolones remain robustly enriched.

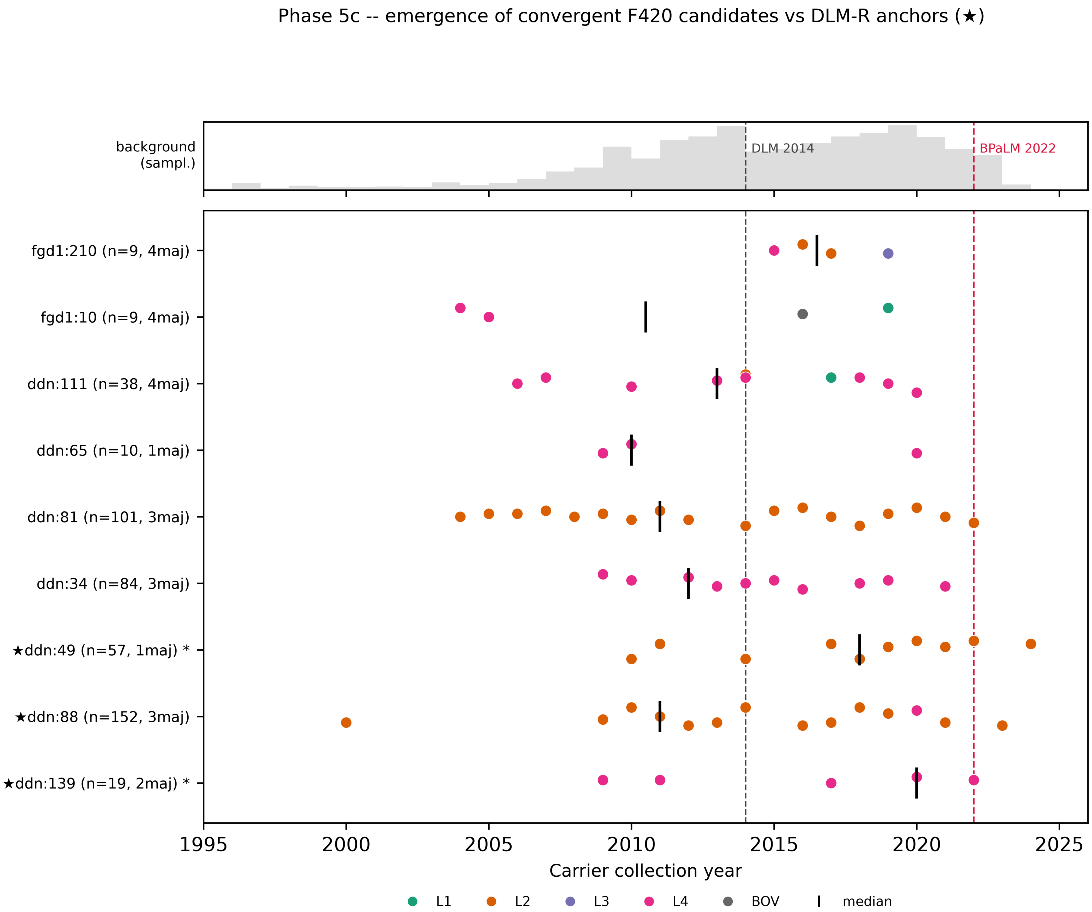

Figure 3: Emergence profile of the convergent F420 candidates versus catalogued delamanid-R anchors (⋆). Each point is a carrier (coloured by major lineage), black bar the median; the grey band shows the dated-sampling background. Convergent candidates are spread across the timeline with pre-2014 carriers (standing), whereas the emergent anchors *ddn* L49P and W139* (marked ∗) sit significantly to the right (drug era). Dashed lines: delamanid clinical use (2014) and BPaLM (2022).

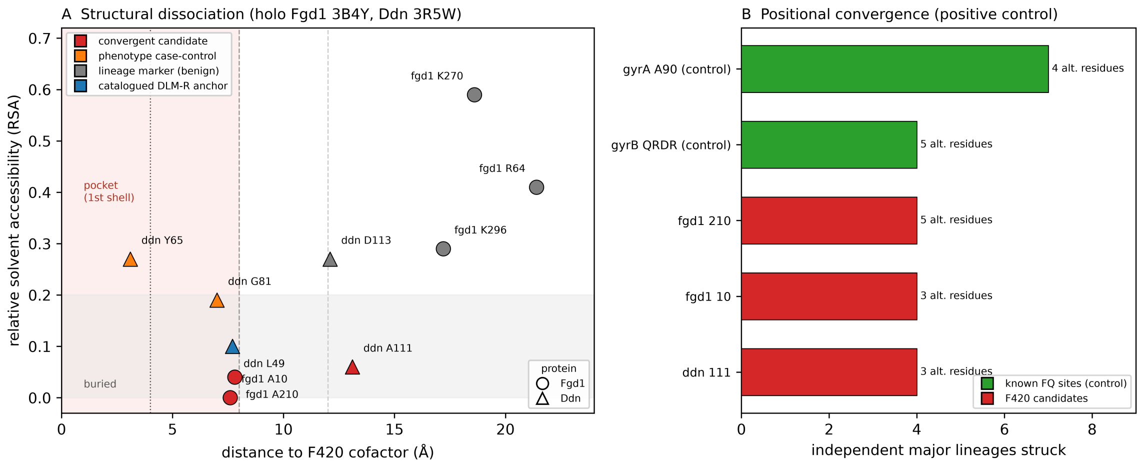

Figure 4: **(A)** Structural dissociation. Distance to the F420 cofactor versus relative solvent accessibility (RSA) for each residue on the holo structures (Fgd1 3B4Y, Ddn 3R5W). Convergent candidates (*fgd1* A10/A210, *ddn* A111), phenotype case-control hits (*ddn* Y65/G81) and the catalogued anchor (*ddn* L49) cluster in the buried pocket zone (<8 Å, RSA <0.20), whereas the lineage markers (*fgd1* R64/K270/K296, *ddn* D113) are distal (12–21 Å) and surface-exposed. **(B)** Positional-convergence positive control: the number of independent major lineages struck (bars) and distinct alternative residues (annotation) for the known fluoroquinolone sites (*gyrA* A90, *gyrB* QRDR) and the F420 candidates. The detector recovers the established fluoroquinolone convergence at a level comparable to the F420 hits.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used Claude Code (Anthropic), an AI-agent software environment powered by the Claude Sonnet 5 model, for in silico data retrieval, statistical analysis, literature verification and drafting assistance, as described in the Materials and Methods. After using this tool, the author reviewed, independently verified against primary sources, and edited the content as necessary, and takes full responsibility for the content of this publication.

References

[1] World Health Organization. Rapid communication: key changes to the treatment of drug-resistant tuberculosis. Technical report, World Health Organization, Geneva, 2022.

[2] Francesca Conradie, Andreas H Diacon, Nosipho Ngubane, Pauline Howell, Daniel Everitt, Angela M Crook, Carl M Mendel, Erica Egizi, Joanna Moreira, Juliano Timm, Timothy D McHugh, Genevieve H Wills, Anna Bateson, Robert Hunt, et al. Treatment of Highly Drug-Resistant Pulmonary Tuberculosis. *New England Journal of Medicine*, 2020. doi: 10.1056/NEJMoa1901814.

[3] Francesca Conradie, Tatevik R Bagdasaryan, Sergey Borisov, Pauline Howell, Lali Mikiashvili, Nosipho Ngubane, Anastasia Samoilova, Sergey Skornykova, Elena Tudor, Ebrahim Variava, Petr Yablonskiy, Daniel Everitt, Genevieve H Wills, et al. Bedaquiline-Pretomanid-Linezolid Regimens for Drug-Resistant Tuberculosis. *New England Journal of Medicine*, 2022. doi: 10.1056/NEJMoa2119430.

[4] Catherine Berry, Philipp du Cros, Katherine Fielding, Suzanne Gajewski, Emil Kazounis, Timothy D McHugh, Corinne Merle, Ilaria Motta, David A J Moore, and Bern-Thomas Nyang'wa. TB-PRACTECAL: study protocol for a randomised controlled phase II-III trial of regimens containing bedaquiline and pretomanid. *Trials*, 2022. doi: 10.1186/s13063-022-06331-8.

[5] Jodie A Schildkraut, Debra Flood, Derek J Sloan, Norbert Heinrich, Shannon Fox, Christoph Lange, Charles Wells, and Simon Tiberi. Rising bedaquiline resistance: exploring potential causes and next steps. *European Respiratory Journal*, 2026. doi: 10.1183/13993003.01531-2025.

[6] Francesca Saluzzo, Seda Yerlikaya, Susan Elizabeth Dorman, Kathleen Eisenach, Maha Reda Farhat, Nazir Ismail, Christian Lienhardt, Timothy M Walker, Fatima Jones, Ana Yepes, Adam Penn-Nicholson, Morten Ruhwald, Bavesh Kana, et al. Future-proofing tuberculosis therapy: framework for concurrent drug and resistance testing development. *The Lancet Infectious Diseases*, 2026. doi: 10.1016/S1473-3099(25)00547-X.

[7] Stephanie Petrella, Emmanuelle Cambau, Aurelie Chauffour, Koen Andries, Vincent Jarlier, and Wladimir Sougakoff. Genetic basis for natural and acquired resistance to the

diarylquinoline R207910 in mycobacteria. *Antimicrobial Agents and Chemotherapy*, 2006. doi: 10.1128/AAC.00244-06.

[8] Anil Koul, Najoua Dendouga, Karen Vergauwen, Brenda Molenberghs, Luc Vranckx, Rudy Willebrords, Zorica Ristic, Holger Lill, Ismet Dorange, Jerome Guillemont, Dirk Bald, and Koen Andries. Diarylquinolines target subunit c of mycobacterial ATP synthase. *Nature Chemical Biology*, 2007. doi: 10.1038/nchembio884.

[9] Koen Andries, Cristina Villellas, Nele Coeck, Kim Thys, Tom Gevers, Luc Vranckx, Nacer Lounis, Bouke C de Jong, and Anil Koul. Acquired resistance of Mycobacterium tuberculosis to bedaquiline. *PLoS ONE*, 2014. doi: 10.1371/journal.pone.0102135.

[10] Ruben C Hartkoorn, Swapna Uplekar, and Stewart T Cole. Cross-resistance between clofazimine and bedaquiline through upregulation of MmpL5 in Mycobacterium tuberculosis. *Antimicrobial Agents and Chemotherapy*, 2014. doi: 10.1128/AAC.00037-14.

[11] Nabila Ismail, Remco P H Peters, Nazir A Ismail, and Shaheed V Omar. Clofazimine Exposure In Vitro Selects Efflux Pump Mutants and Bedaquiline Resistance. *Antimicrobial Agents and Chemotherapy*, 2019. doi: 10.1128/AAC.02141-18.

[12] Nabila Ismail, Nazir A Ismail, Shaheed V Omar, and Remco P H Peters. In Vitro Study of Stepwise Acquisition of rv0678 and atpE Mutations Conferring Bedaquiline Resistance. *Antimicrobial Agents and Chemotherapy*, 2019. doi: 10.1128/AAC.00292-19.

[13] Qinglong Guo, Jing Bi, Qiao Lin, Taosheng Ye, Zhongyuan Wang, Zhaoqin Wang, Lei Liu, and Guoliang Zhang. Whole Genome Sequencing Identifies Novel Mutations Associated With Bedaquiline Resistance in Mycobacterium tuberculosis. *Frontiers in Cellular and Infection Microbiology*, 2022. doi: 10.3389/fcimb.2022.807095.

[14] Stuti Ghosh, Sudipto Bhattacharjee, and Sudipto Saha. Machine learning method for the prediction of Bedaquiline-resistant Mycobacterium tuberculosis. *Life Science Alliance*, 2026. doi: 10.26508/lsa.202503539.

[15] Ujjini H Manjunatha, Helena Boshoff, Cynthia S Dowd, Liang Zhang, Thomas J Albert, Jason E Norton, Lacy Daniels, Thomas Dick, Siew Siew Pang, and Clifton E Barry. Identification of a nitroimidazo-oxazine-specific protein involved in PA-824 resistance in Mycobacterium tuberculosis. *Proceedings of the National Academy of Sciences USA*, 2006. doi: 10.1073/pnas.0508392103.

[16] Ghader Bashiri, Christopher J Squire, Edward N Baker, and Nicole J Moreland. Expression, purification and crystallization of Mycobacterium tuberculosis FGD1 (Rv0407) using a Mycobacterium smegmatis expression system. *Protein Expression and Purification*, 2007. doi: 10.1016/j.pep.2007.01.014.

[17] K P Choi, T B Bair, Y M Bae, and L Daniels. Use of transposon Tn5367 mutagenesis and a nitroimidazopyran-based selection system to demonstrate a requirement for fbiA and fbiB in coenzyme F420 biosynthesis by Mycobacterium bovis BCG. *Journal of Bacteriology*, 2001. doi: 10.1128/JB.183.24.7058-7066.2001.

[18] Kwang-Pil Choi, Nathan Kendrick, and Lacy Daniels. Demonstration that fbiC is required by Mycobacterium bovis BCG for coenzyme F420 and FO biosynthesis. *Journal of Bacteriology*, 2002. doi: 10.1128/JB.184.9.2420-2428.2002.

[19] Amit Sharma, Poulomi Chatterjee, and Shashank Shastry. Delamanid or pretomanid for treatment of multidrug resistant tuberculosis - spoilt for choice? *Indian Journal of Tuberculosis*, 2026. doi: 10.1016/j.ijtb.2025.09.005.

[20] Jason E Cummings, Lisa K Woolhiser, Vincent Guglielmi, Ashely Romano, Samantha Pauly, Matt Zimmerman, Firat Kaya, Nicholas D Walter, Gregory T Robertson, and Richard A Slayden. Pretomanid vs delamanid in a bedaquiline-linezolid regimen: efficacy in a high-burden tuberculosis mouse model. *Antimicrobial Agents and Chemotherapy*, 2026. doi: 10.1128/aac.01953-25.

[21] Qian Zhang, Peng Xu, Xuecong Zhang, Rui Pi, Yiwang Chen, Howard Takiff, and Qian Gao. Spontaneous mutations to delamanid and pretomanid resistance in Mycobacterium tuberculosis: SNPs confer high-level resistance. *International Journal of Antimicrobial Agents*, 2026. doi: 10.1016/j.ijantimicag.2026.107789.

[22] Md Mahmudul Islam, Md Zahid Hasan, Md Touki Tahamid Tusar, Md Yeamin Hossain, Md Motaher Hossain, Md Abdulla Al Jubayed, et al. Molecular Mechanisms of Resistance and Treatment Efficacy of Delamanid Against Mycobacterium tuberculosis: A Systematic Review. *Health Science Reports*, 2026. doi: 10.1002/hsr2.72481.

[23] Shalini Saxena and Lalitha Guruprasad. Elucidating the structural basis of Fgd1-mediated resistance to Delamanid in Mycobacterium tuberculosis. *Journal of Molecular Graphics and Modelling*, 2026. doi: 10.1016/j.jmgm.2026.109382.

[24] Gourav Chakraborty, Mahima Sudhir Kolpe, I V Ambily Nath, Avlokita Tiwari, Praapti Jayaswal, and Niladri Patra. Computational structure-guided approach to simulate delamanid and pretomanid binding to mycobacterial F420 redox cycling proteins: identification of key determinants of resistance. *Journal of Biomolecular Structure and Dynamics*, 2025. doi: 10.1080/07391102.2025.2498621.

[25] Faridath Massou, Klaas Dewaele, Anderson Gnamy, Ramata Balde, Wim Mulders, Conor J Mehan, Bouke C de Jong, Dissou Affolabi, Oren Tzfadia, and Leen Rigouts. No phenotypic resistance observed for most group-3 and -4 variants in Mycobacterium tuberculosis genes related to bedaquiline, clofazimine, delamanid, and pretomanid in a Central and West African context. *Microbiology Spectrum*, 2026. doi: 10.1128/spectrum.02005-26.

[26] World Health Organization. Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance, second edition. Technical report, World Health Organization, Geneva, 2023.

[27] Martina L Reichmuth, Rico Hömke, Kathrin Zürcher, Peter Sander, Anchalee Avihingsanon, Jimena Collantes, Chloé Loiseau, Sonia Borrell, Miriam Reinhard, Robert J Wilkinson, Marcel Yotebieng, Lukas Fenner, Erik C Böttger, Sebastien Gagneux, Matthias Egger, and

Peter M Keller. Natural polymorphisms in mycobacterium tuberculosis conferring resistance to delamanid in drug-naive patients. *Antimicrobial Agents and Chemotherapy*, 64 (11):e00513–20, 2020. doi: 10.1128/AAC.00513-20.

[28] Paula J Gómez-González, Joao Perdigao, Pedro Gomes, Zully M Puyen, David Santos-Lazaro, Gary Napier, Martin L Hibberd, Miguel Viveiros, Isabel Portugal, Susana Campino, Jody E Phelan, and Taane G Clark. Genetic diversity of candidate loci linked to mycobacterium tuberculosis resistance to bedaquiline, delamanid and pretomanid. *Scientific Reports*, 11(1):19431, 2021. doi: 10.1038/s41598-021-98862-4.

[29] Anna Bateson, Julio Ortiz Canseco, Timothy D McHugh, Adam A Witney, Silke Feuerriegel, Matthias Merker, Thomas A Kohl, Christian Utpatel, Stefan Niemann, Sönke Andres, Katharina Kranzer, Florian P Maurer, Arash Ghodousi, Emanuele Borroni, Daniela Maria Cirillo, Maria Wijkander, Juan C Toro, Ramona Groenheit, Jim Werngren, Diana Machado, Miguel Viveiros, Robin M Warren, Frederick Sirgel, Anzaan Dippenaar, Claudio U Köser, Eugene Sun, and Juliano Timm. Ancient and recent differences in the intrinsic susceptibility of mycobacterium tuberculosis complex to pretomanid. *Journal of Antimicrobial Chemotherapy*, 77(6):1685–1693, 2022. doi: 10.1093/jac/dkac070.

[30] Praharshinie Rupasinghe, Rabab Reenaers, Jens Vereecken, Wim Mulders, Sari Cogneau, Matthias Merker, Stefan Niemann, Shaheed Vally Omar, Leen Rigouts, Claudio U Köser, Tom Decroo, and Bouke C de Jong. Refined understanding of the impact of the mycobacterium tuberculosis complex diversity on the intrinsic susceptibility to pretomanid. *Microbiology Spectrum*, 12(3):e00070–24, 2024. doi: 10.1128/spectrum.00070-24.

[31] Emmanuel Rivière, Lennert Verboven, Anzaan Dippenaar, Sander Goossens, Elise De Vos, Elizabeth Streicher, Bart Cuypers, Kris Laukens, Fathia Ben-Rached, Timothy C Rodwell, Arnab Pain, Robin M Warren, Tim H Heupink, and Annelies Van Rie. Variants in bedaquiline-candidate-resistance genes: Prevalence in bedaquiline-naive patients, effect on mic, and association with mycobacterium tuberculosis lineage. *Antimicrobial Agents and Chemotherapy*, 66(7):e00322–22, 2022. doi: 10.1128/aac.00322-22.

[32] Patrick Beckert, Elisabeth Sanchez-Padilla, Matthias Merker, Viola Dreyer, Thomas A Kohl, Christian Utpatel, Claudio U Köser, Ivan Barilar, Nazir Ismail, Shaheed Vally Omar, Marisa Klopper, Robin M Warren, Harald Hoffmann, et al. MDR M. tuberculosis outbreak clone in Eswatini missed by Xpert has elevated bedaquiline resistance dated to the pre-treatment era. *Genome Medicine*, 2020. doi: 10.1186/s13073-020-00793-8.

[33] Camus Nimmo, Arturo Torres Ortiz, Cedric C S Tan, Juanita Pang, Mislav Acman, James Millard, Nesri Padayatchi, Alison D Grant, Max O'Donnell, Alex Pym, Ola B Brynildsrud, Vegard Eldholm, Louis Grandjean, Xavier Didelot, François Balloux, and Lucy van Dorp. Detection of a historic reservoir of bedaquiline/clofazimine resistance-associated variants in mycobacterium tuberculosis. *Genome Medicine*, 16(1):34, 2024. doi: 10.1186/s13073-024-01289-5.

[34] Silke Feuerriegel, Claudio U Köser, Davide Baù, Sabine Rüsch-Gerdes, David K Summers, John A C Archer, Marc A Marti-Renom, and Stefan Niemann. Impact of fgd1 and ddn diversity in mycobacterium tuberculosis complex on in vitro susceptibility to pa-824. *Antimicrobial Agents and Chemotherapy*, 55(12):5718–5722, 2011. doi: 10.1128/AAC.05500-11.

[35] Derek Conkle-Gutierrez, Bria M Gorman, Nachiket Thosar, Afif Elghraoui, Samuel J Modlin, and Faramarz Valafar. Widespread loss-of-function mutations implicating preexisting resistance to new or repurposed anti-tuberculosis drugs. *Drug Resistance Updates*, 77:101156, 2024. doi: 10.1016/j.drup.2024.101156.

[36] Arturo Torres Ortiz, Jorge Coronel, Julia Rios Vidal, Cesar Bonilla, David A J Moore, Robert H Gilman, Francois Balloux, Onn Min Kon, Xavier Didelot, and Louis Grandjean. Genomic signatures of pre-resistance in mycobacterium tuberculosis. *Nature Communications*, 12(1):7312, 2021. doi: 10.1038/s41467-021-27616-7.

[37] Juliano Timm, Anna Bateson, Priya Solanki, Ana Paleckyte, Adam A Witney, Sylvia A D Rofael, Stella Fabiane, Morounfolu Olugbosi, Timothy D McHugh, and Eugene Sun. Baseline and acquired resistance to bedaquiline, linezolid and pretomanid, and impact on treatment outcomes in four tuberculosis clinical trials containing pretomanid. *PLOS Global Public Health*, 3(10):e0002283, 2023. doi: 10.1371/journal.pgph.0002283.

[38] I Flores, M A A Tujan, R Basilio, T J R Dizon, D R Lim, L T Reyes, R V A G Reyes, S Rojas, A K Valencia, J B Capin, V Mirtskhulava, et al. High efficacy of BPaL among patients infected with Mycobacterium tuberculosis lineage 1 in the Philippines. *IJTLD Open*, 2026. doi: 10.5588/ijtldopen.25.0677.

[39] Valerie F A March, Michaela Zwyer, Chloé Loiseau, Daniela Brites, Galo A Goig, Selim Bouaouina, Anna Doetsch, Miriam Reinhard, Sevda Kalkan, Sebastien Gagneux, and Sonia Borrell. Variability in intrinsic drug tolerance in Mycobacterium tuberculosis corresponds with phylogenetic lineage. *Antimicrobial Agents and Chemotherapy*, 2025. doi: 10.1128/aac.00996-25.

[40] Amir Banaei-Esfahani, Sonia Borrell, Andrej Trauner, Sebastian M Gygli, Tige R Rustad, Julia Feldmann, Ludovic C Gillet, Olga T Schubert, Christian Beisel, David R Sherman, Ruedi Aebersold, Sebastien Gagneux, and Ben C Collins. LysG-driven transcriptional network rewiring underlies lineage-specific phenotypes in Mycobacterium tuberculosis. *Nature Communications*, 2026. doi: 10.1038/s41467-026-70539-4.

[41] Gaetan Senelle, Muhammed Rabiu Sahal, Kevin La, Tiphaine Billard-Pomares, Julie Marin, Faiza Mougari, Antoine Bridier-Nahmias, Etienne Carbonnelle, Emmanuelle Cambau, Guislaine Refrégier, Christophe Guyeux, and Christophe Sola. Towards the reconstruction of a global TB history using a new pipeline "TB-Annotator". *Tuberculosis*, 143:102376, 2023. doi: 10.1016/j.tube.2023.102376.

[42] Francesc Coll, Ruth McNerney, José Afonso Guerra-Assunção, Judith R Glynn, João Perdigão, Miguel Viveiros, Isabel Portugal, Arnab Pain, Nigel Martin, and Taane G Clark.

A robust snp barcode for typing mycobacterium tuberculosis complex strains. *Nature Communications*, 5:4812, 2014. doi: 10.1038/ncomms5812.

[43] Joshua Meier, Roshan Rao, Robert Verkuil, Jason Liu, Tom Sercu, and Alexander Rives. Language models enable zero-shot prediction of the effects of mutations on protein function. In *Advances in Neural Information Processing Systems*, volume 34, pages 29287–29303, 2021.

[44] Peter JA Cock, Tiago Antao, Jeffrey T Chang, Brad A Chapman, Cymon J Cox, Andrew Dalke, Iddo Friedberg, Thomas Hamelryck, Frank Kauff, Bartek Wilczynski, and Michiel JL de Hoon. Biopython: freely available python tools for computational molecular biology and bioinformatics. *Bioinformatics*, 25(11):1422–1423, 2009. doi: 10.1093/bioinformatics/btp163.

[45] Matthew Z Tien, Austin G Meyer, Dariya K Sydykova, Stephanie J Spielman, and Claus O Wilke. Maximum allowed solvent accessibilities of residues in proteins. *PLoS ONE*, 8(11): e80635, 2013. doi: 10.1371/journal.pone.0080635.

[46] David J Edwards, Sebastián Duchene, Bernard Pope, and Kathryn E Holt. Snppar: identifying convergent evolution and other homoplasies from microbial whole-genome alignments. *Microbial Genomics*, 7(12):000694, 2021. doi: 10.1099/mgen.0.000694.

[47] Timothy M Walker, Camilla L C Ip, Ruth H Harrell, Jason T Evans, Georgia Kapatai, Martin J Dedicoat, David W Eyre, Daniel J Wilson, Peter M Hawkey, Derrick W Crook, Julian Parkhill, David Harris, A Sarah Walker, Rory Bowden, Philip Monk, E Grace Smith, and Tim E A Peto. Whole-genome sequencing to delineate Mycobacterium tuberculosis outbreaks: a retrospective observational study. *The Lancet Infectious Diseases*, 2013. doi: 10.1016/S1473-3099(12)70277-3.

[48] Shaojun Pei, Wei Yang, Xichao Ou, Bing Zhao, Zeyuan Zhao, Zekun Wang, Zexuan Song, Shengfen Wang, Jingyuan Zhuang, Chen Li, and Yanlin Zhao. High-level primary pretomanid-resistant with ddn in-frame deletion of mycobacterium tuberculosis and its association with lineage 4.5 in china. *ACS Omega*, 10(25):26551–26559, 2025. doi: 10.1021/acsomega.5c00610.

[49] Kamel Djaout, Aurore Dreneau, Léo Faion, Marion Prieri, Maxime Neuville, Alice Boulloy, Alexandre Biela, Rosangela Frita, Marie Devaere, Stephanie Slupek, Rudy Antoine, Ruben C Hartkoorn, et al. Exploiting DprE2 pathway redundancy to enhance potency and overcome resistance to pretomanid in Mycobacterium tuberculosis. *npj Antimicrobials and Resistance*, 2026. doi: 10.1038/s44259-026-00201-y.

[50] Praharshinie Rupasinghe, Nabila Ismail, Wim Mulders, Robin M Warren, Lavania Joseph, Dumisani Ngcamu, Thabisile Gwala, Shaheed V Omar, Jens Vereecken, Bouke C de Jong, Leen Rigouts, Emanuele Borroni, Daniela M Cirillo, Thomas Schön, Paolo Miotto, and Claudio U Köser. In vitro exposure to clofazimine can select for delamanid and pretomanid resistance in Mycobacterium tuberculosis. *Antimicrobial Agents and Chemotherapy*, 2025. doi: 10.1128/aac.01113-25.

[51] Nazanin Ahmad Khosravi, Azar Dokht Khosravi, Mohammad Hashemzadeh, Mohammad Savari, and Morteza Saki. Investigation of delamanid, bedaquiline, and linezolid resistance rates and related gene mutations in multidrug-resistant Mycobacterium tuberculosis in regional tuberculosis reference laboratories of Iran. *BMC Microbiology*, 2025. doi: 10.1186/s12866-025-04213-y.

[52] Mikael Mansjö, Oskar Karlsson Lindsjö, Christina Grönfors Seeth, Ramona Groenheit, and Jim Werngren. The ddn trp20stop mutation and its association with lineage 4.5 and resistance to delamanid and pretomanid in mycobacterium tuberculosis. *Antimicrobial Agents and Chemotherapy*, 66(12):e01026–22, 2022. doi: 10.1128/aac.01026-22.

[53] Cristina Villellas, Frederik Stevenaert, Bart Remmerie, and Koen Andries. Sub-mic levels of bedaquiline and clofazimine can select mycobacterium tuberculosis mutants with increased mic. *Antimicrobial Agents and Chemotherapy*, 68(4):e01275–23, 2024. doi: 10.1128/aac.01275-23.

[54] Seungmo Kim, Seung Heon Lee, Gisu Kang, Gyeong In Lee, Hyeon-Su Kim, Jeong Seong Yang, Youngsuk Park, Byoung Oh Hwang, and Hyejin Kim. Functional analysis of genetic mutations in ddn and fbia linked to delamanid resistance in rifampicin-resistant mycobacterium tuberculosis. *Tuberculosis*, 152:102630, 2025. doi: 10.1016/j.tube.2025.102630.

[55] Dalin Rifat, Si-Yang Li, Thomas Ioerger, Keshav Shah, Jean-Philippe Lanoix, Jin Lee, Ghader Bashiri, James Sacchettini, and Eric Nuermberger. Mutations in fbid (rv2983) as a novel determinant of resistance to pretomanid and delamanid in mycobacterium tuberculosis. *Antimicrobial Agents and Chemotherapy*, 65(1):e01948–20, 2020. doi: 10.1128/AAC.01948-20.

[56] Fuzhen Zhang, Shanshan Li, Shuan Wen, Tingting Zhang, Yuanyuan Shang, Fengmin Huo, Yi Xue, Ling Li, and Yu Pang. Comparison of in vitro susceptibility of mycobacteria against pa-824 to identify key residues of ddn, the deazaflavin-dependent nitroreductase from mycobacterium tuberculosis. *Infection and Drug Resistance*, 13:815–822, 2020. doi: 10.2147/IDR.S240716.

[57] Ravi Singh, Muhammed Shaheer, and M Elizabeth Sobhia. Molecular dynamic assisted investigation on impact of mutations in deazaflavin dependent nitroreductase against pretomanid: a computational study. *Journal of Biomolecular Structure & Dynamics*, 41(10): 4421–4443, 2022. doi: 10.1080/07391102.2022.2069156.

[58] Brendon M Lee, Liam K Harold, Deepak V Almeida, Livnat Afriat-Jurnou, Htin Lin Aung, Brian M Forde, Kiel Hards, Sacha J Pidot, F Hafna Ahmed, A Elaaf Mohamed, Matthew C Taylor, Nicholas P West, Timothy P Stinear, Chris Greening, Scott A Beatson, Eric L Nuermberger, Gregory M Cook, and Colin J Jackson. Predicting nitroimidazole antibiotic resistance mutations in mycobacterium tuberculosis with protein engineering. *PLoS Pathogens*, 16(2):e1008287, 2020. doi: 10.1371/journal.ppat.1008287.

[59] Chendi Zhu, Tingting Yang, Jinfeng Yin, Hui Jiang, Howard E Takiff, Qian Gao, Qingyun Liu, and Weimin Li. The global success of mycobacterium tuberculosis modern beijing

family is driven by a few recently emerged strains. *Microbiology Spectrum*, 11(4):e03339–22, 2023. doi: 10.1128/spectrum.03339-22.

[60] Chloé Loiseau, Etthel M Windels, Sebastian M Gygli, Levan Jugheli, Nino Maghradze, Daniela Brites, Amanda Ross, Galo Goig, Miriam Reinhard, Sonia Borrell, Andrej Trauner, Anna Dötsch, Rusudan Aspindzelashvili, Rebecca Denes, Klaus Reither, Christian Beisel, Nestani Tukvadze, Zaza Avaliani, Tanja Stadler, and Sebastien Gagneux. The relative transmission fitness of multidrug-resistant mycobacterium tuberculosis in a drug resistance hotspot. *Nature Communications*, 14(1):1988, 2023. doi: 10.1038/s41467-023-37719-y.

[61] Jean-Philippe Rasigade, Maxime Barbier, Oana Dumitrescu, Catherine Pichat, Gérard Carret, Anne-Sophie Ronnaux-Baron, Ghislaine Blasquez, Christine Godin-Benhaim, Sandrine Boisset, Anne Carricajo, Véronique Jacomo, Isabelle Fredenucci, Michèle Pérouse de Montclos, Jean-Pierre Flandrois, Florence Ader, Philip Supply, Gérard Lina, and Thierry Wirth. Strain-specific estimation of epidemic success provides insights into the transmission dynamics of tuberculosis. *Scientific Reports*, 7:45326, 2017. doi: 10.1038/srep45326.